

Ownership Consolidation and Product Characteristics: A Study of the US Daily Newspaper Market

Ying Fan (2013). *American Economic Review*.

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Motivation

- Standard merger analysis omits the role of **product characteristics**
- Underestimate welfare loss for consumers / gain for firms

This paper:

- Examine the US daily newspaper market
 - Endogenize: price & editorial decisions
- Counterfactual analysis
 - A banned merger of 2 newspapers in Minneapolis-St. Paul metro area
 - Duopoly and triopoly mergers

Model Overview: Features of the Newspaper Market

- 3 types of agents: readers, advertisers, publishers
 1. Readers: subscribe to ≤ 2 newspapers per year
 2. Advertiser: demand for advertising space
 3. Publisher: maximize (joint) profit from subscription & advertising revenue
- Newspapers have differentiated characteristics
 - Endogenous: annual subscription price, content quality, local news ratio, variety

Model - Reader's Demand

Utility of subscribing to newspaper j , for household i in county c in year t :

$$u_{ijct} = p_{jt}\alpha + \underbrace{\mathbf{x}_{jt}}_{\text{endo. chars.}} \beta_{ict} + \underbrace{\mathbf{y}_{jct}}_{\text{exo. chars.}} \psi + \underbrace{\mathbf{z}_{ct}}_{\text{demographics}} \varphi + \xi_{jct} + \varepsilon_{ijt}$$

Random coefficient for k -th endogenous characteristics:

$$\beta_{kict} = \beta_k + \mathbf{z}_{ct}\boldsymbol{\theta}_k + \sigma_k S_{kict}$$

Outside choice (decreasing interest in reading newspapers)

$$u_{i0t} = (t - t_0)\rho + \varepsilon_{i0t}$$

Product Characteristics

- Endogenous $\mathbf{x}_{jt}$:

1. Content quality: $x_{1jt} = x_{11jt} + \varpi_2 x_{12jt} + \varpi_3 x_{13jt}$

- x_{11jt} : nonadvertising space
- x_{12jt} : # staff for opinion section
- x_{13jt} : # reporters

2. Local news ratio

3. Variety: HHI of share of staff in each section

- Exogenous $\mathbf{y}_{jct}$:

- # households in the market, morning edition, local newspaper, distance to newspaper's home county

Reader's multiple discrete choices problem

Assume each household subscribes to **at most 2 newspapers**

Market penetration = the probability of subscribing to newspaper j :

1. j is the best choice: $\Pr(u_{ijct} \geq \max_{h=0, \dots, J_{ct}} u_{ihct})$

2. j is the second best:

$$\sum_{j' \neq j} \Pr \left(\underbrace{u_{ij'ct} \geq u_{ijct} \geq \max_{h \neq j'} u_{ihct}}_{j \text{ ranked 2nd}}, \underbrace{u_{ijct} - \kappa}_{\text{gains from 2nd}} \geq u_{i0ct} \right)$$

where

- J_{ct} : set of newspapers in county c in year t
- κ : disutility for subscribing to the second newspaper

Model - Advertiser's Demand

Consider a Cobb-Douglas form of advertising demand:

$$a(r_{jt}, q_{jt}, H_{jt}; \eta, \boldsymbol{\lambda}) = e^{\eta} H_{jt}^{\lambda_0} q_{jt}^{\lambda_1} r_{jt}^{\lambda_2}$$

$$\log a_{jt} = \eta + \lambda_0 \log H_{jt} + \lambda_1 \log q_{jt} + \lambda_2 \log r_{jt} + \iota_{jt}$$

- H_{jt} : # of households in newspaper j 's circulation area
- q_{jt} : total circulation
- r_{jt} : advertising rate

Model - Supply

Overlapping are prevalent. We have to define **markets** where newspapers strategically interact with each other.

1. National newspapers are taken as exogenous: *WSJ, NYT, USA Today*
2. For each newspaper, drop trivial counties that circulate almost no copies
3. Economies of scope only exist in the home market (defined by *MSA*)

Newspaper's sources of revenue:

1. Subscription revenue
2. Preprint advertising revenue
3. Display advertising revenue

Supply - Publisher's Profit Maximization Problem

In a two-stage game, a publisher j decides on:

1. Make editorial decisions (choose $\mathbf{x}_{jt}$)
2. Set price p_{jt} and advertising rate r_{jt}

to maximize profit:

$$\pi_j^I(\mathbf{x}) = \underbrace{\pi_j^{II}(\mathbf{p}^*(\mathbf{x}), \mathbf{r}^*(\mathbf{x}); \mathbf{x})}_{\text{2nd stage profit}} - \underbrace{\text{fc}(\mathbf{x}_j, \boldsymbol{\nu}_j; \boldsymbol{\tau})}_{\text{quadratic fixed costs}}$$

where

- $\mathbf{p}^*(\mathbf{x})$: equilibrium price vector
- $\mathbf{r}^*(\mathbf{x})$: equilibrium advertising rate vector
- $\boldsymbol{\nu}_j$: unobservable cost shocks

Supply - Cost & Profit Functions

- Average costs per copy:

$$ac_j^{(q)} = \left(\gamma_1 + \gamma_2 f_j + \gamma_3 \underbrace{(x_{1j} + a_j)}_{\text{pages}} \right) \underbrace{\log(Q_j)^{\gamma_4}}_{\text{decreasing in total qty}} + \omega_j$$

- Ad profit:

$$\left(r_j - \underbrace{(1 + 1/\lambda_2)(\bar{\zeta} + \zeta_j)}_{\text{MC of ad sales}} \right) \cdot \underbrace{a_j}_{\text{convex in qty}}$$

- Preprint profit: $\mu_1 q_j + \frac{1}{2} \mu_2 q_j^2$

Intuitively, scale economies may come from **lower average costs & higher ad demand**.

Data

- 1997-2005
- Reader side:
 - county circulation, annual subscription price
- Advertiser side:
 - annual display ad lineage, display ad rates
- Publisher side:
 - 422 newspapers per year
 - Joint Operation Agreement

Summary Statistics

TABLE 1—SUMMARY STATISTICS OF PLAYER NEWSPAPERS

	Mean	Median	SD	Min	Max	Observations
Total circulation	19,117	9,322	32,739	1,132	468,483	5,843 ^a
Annual display ad lineage (column inch)	1,362,154	1,168,452	760,828	123,124	4,337,941	422
Price of newspapers (\$)	101.24	96.61	33.75	13.19	365.31	5,843
Display advertising rate (\$/column inch)	23.50	12.91	36.16	2.45	520.29	5,843
Frequency (issues/52 weeks)	308.16	312.00	53.70	208.00	364.00	5,843
Pages (pages/issue)	27.06	22.86	16.47	8.00	167.43	5,843
Opinion staff	1.36	0.33	3.05	0	42.42	5,843
Reporters	5.31	2.00	12.44	0	210.00	5,843
Local news ratio (percent)	13.69	13.89	10.16	0	62.50	5,843
Variety	69.80	75.06	17.50	0	88.63	5,843
Market penetration (percent)	33.70	32.82	16.63	0.56	97.08	11,203 ^b
Distance to home county (100km)	0.26	0	0.38	0	3.95	11,203

^aThese observations are at the newspaper/year level.
^bThese observations are at the newspaper/county/year level.

TABLE 2—SUMMARY STATISTICS: COUNTY DEMOGRAPHICS

	Mean	Median	SD	Min	Max	Observations
Percent of population over 25 with bachelor’s degree or higher	16.88	15.05	6.95	5.64	60.48	8,947 ^a
Median income (\$1,000)	34.17	32.82	7.09	16.36	76.31	
Median age	36.56	36.70	3.73	20.70	54.30	
Urbanization (percent)	48.80	50.50	26.05	0	100	
Number of households	28,550	14,876	47,949	710	858,719	

^aThese observations are at the county/year level.

Estimation

Apply GMM. Moment conditions are from:

- Demand system: (1) Newspaper demand; (2) display ads demand
- Equilibrium conditions: FOC of the profit maximization problem
 - (3) advertising rate
 - (4) price
 - (5) editorial decisions (endogenous characteristics)
- Excluded instrument: the demographics in the market of j 's competitors (excluding j 's own market)
 - Relevant for the decision of j 's competitors, hence j 's decision
 - Exclusion restriction: only affect j 's demand through its competitors
 - Education, income, age, urbanization

Parameter Estimates

	Parameter	Estimate	SE					
Utility				Display advertising demand				
	Price (\$100)	α	−1.411***	0.030	Ad market size	λ_0	0.043**	0.026
	log(1 + content quality)	β_1	0.772***	0.017	Total circulation	λ_1	1.673***	0.039
	log(1 + local news)	β_2	−0.002***	0.001	Ad rate	λ_2	−1.195***	0.029
	log(1 + variety)	β_3	0.013***	0.001	Constant	ϕ	−1.625***	0.269
	Weight on news hole in content quality	1			Average cost of circulation			
	Weight on opinion in content quality	ϖ_2	0.466*	0.301	Constant	γ_1	−82.315	145.240
	Weight on reporters in content quality	ϖ_3	6.264***	0.322	Frequency	γ_2	1.768***	0.112
	log(1 + local news), education	θ_1	0.003***	0.000	Pages in a year	γ_3	0.022***	0.004
	log(1 + local news), median age	θ_2	0.004***	0.001	Economies of scale/scope	γ_4	−0.448***	0.045
log(1 + local news), standard deviation	σ	0.007	0.016	Marginal cost of ad sales	$\bar{\zeta}$	11.775***	0.518	

Figure 1: Estimation Results

There are economies of scale ($\hat{\lambda}_1 > 1, \hat{\gamma}_4 < 0$).

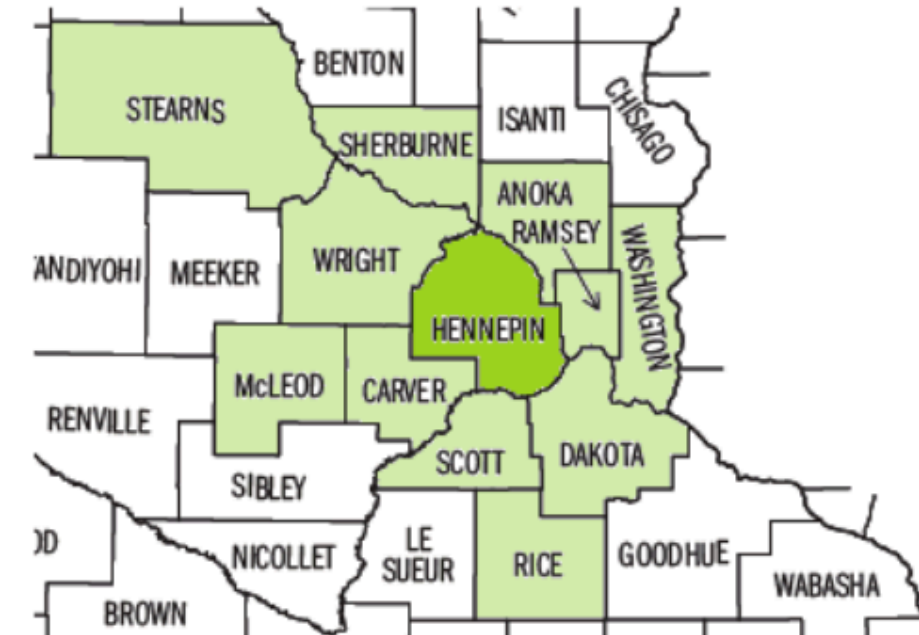
$\kappa = 1.919 \rightarrow$ Duplicate readership is negligible in about 45% of counties with multiple newspapers

Counterfactual Analyses

Merger of *Minneapolis Star Tribune* & *St. Paul Pioneer*

- 2006: The publisher of the *Star* acquired the publisher of the *Pioneer*
- 3 months later blocked by Department of Justice
- The *Pioneer* was sold to a third party that owned no newspapers before the acquisition
- Other publishers: *Faribault*, *St. Cloud*, *Stillwater*
- What if the merger was allowed?

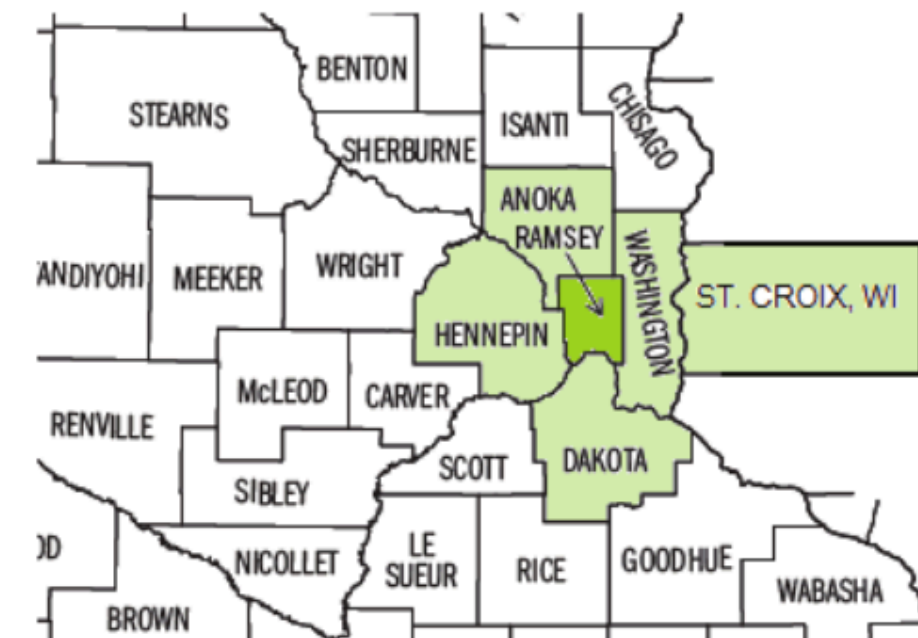
Panel A. Market of the *Star Tribune*



Market of the *Star Tribune*

Hennepin (Home County),
Anoka, Carver, Dakota, McLeod,
Ramsey, Rice, Scott, Sherburne,
Stearns, Washington, Wright

Panel B. Market of the *Pioneer Press*



Market of the *Pioneer Press*

Ramsey (Home County),
Anoka, Dakota, Hennepin,
St. Croix, Washington

Newspaper Designated Market

Without Characteristic Adjustment

TABLE 5—EFFECTS OF OWNERSHIP CONSOLIDATION OF THE *STAR*
AND THE *PIONEER* WITHOUT CHARACTERISTIC ADJUSTMENT

		Star	Pioneer	Faribault	St. Cloud	Stillwater
Price (\$/year)	before	172.79	171.51	111.31	150.07	78.33
	after	175.98	179.52	111.32	149.95	75.03
	change	3.19	8.01	0	−0.12	−3.30
Ad rate (\$/column inch)	before	230.88	153.08	12.37	44.15	11.13
	after	227.00	147.07	12.39	44.19	11.25
	change	−3.87	−6.00	0.02	0.03	0.12
Circulation	before	317,337	159,864	6,384	24,578	3,341
	after	310,148	148,519	6,434	24,667	3,644
	change	−7,189	−11,345	50	89	303

Effects of Consolidation **without** Characteristic Adjustment

With Characteristic Adjustment

TABLE 6—EFFECTS OF OWNERSHIP CONSOLIDATION OF THE *STAR*
AND THE *PIONEER* WITH CHARACTERISTIC ADJUSTMENT

		Star	Pioneer	Faribault	St. Cloud	Stillwater
Content quality index	before	788.49	474.29	7.00	65.28	0.70
	after	771.78	422.59	7.17	66.26	0.31
	change	−16.72	−51.70	0.17	0.98	−0.40
Local news (percent)	before	22.00	27.48	14.29	35.42	0
	after	21.15	23.88	14.47	35.60	0.05
	change	−0.85	−3.60	0.18	0.18	0.05
Variety	before	83.38	82.07	50.00	74.50	0
	after	81.79	74.61	50.35	75.01	0
	change	−1.58	−7.46	0.35	0.51	0
Price (\$/year)	before	172.79	171.51	111.31	150.07	78.33
	after	175.39	178.83	111.26	149.64	87.41
	change	2.59	7.32	−0.05	−0.43	9.08
Ad rate (\$/column inch)	before	230.88	153.08	12.37	44.15	11.13
	after	227.09	144.4	12.42	44.29	10.83
	change	−3.79	−8.68	0.05	0.13	−0.3
Circulation	before	317,337	159,864	6,384	24,578	3,341
	after	310,223	140,635	6,518	24,939	2,597
	change	−7,114	−19,229	134	361	−744

Effects of Consolidation **with** Characteristic Adjustment

Reader, Advertiser, and Publisher Surplus

TABLE 7—WELFARE EFFECTS OF OWNERSHIP CONSOLIDATION OF THE *STAR* AND THE *PIONEER*

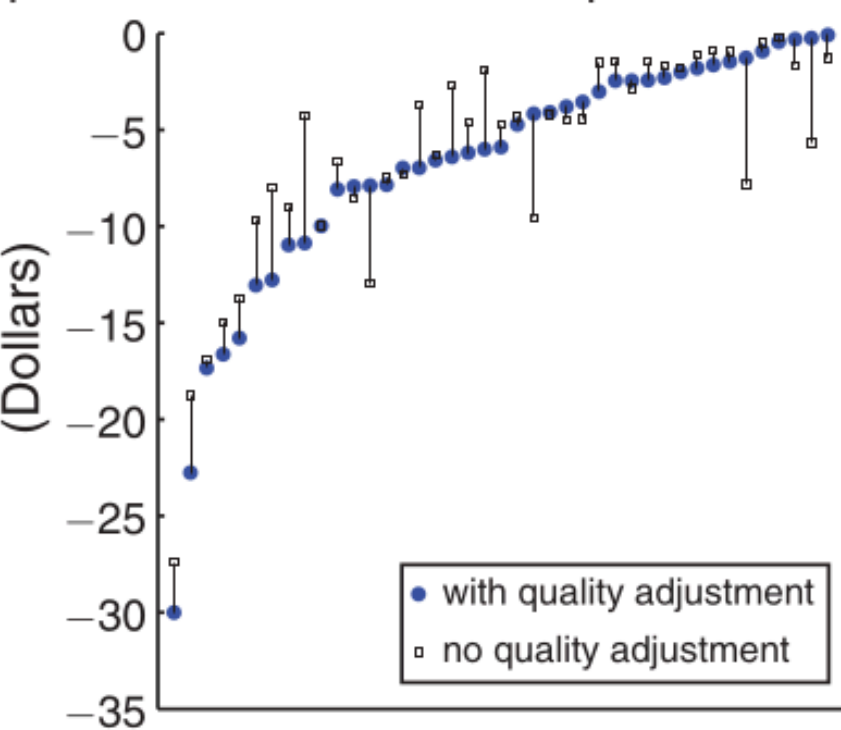
	Without characteristic adjustment	With characteristic adjustment
Change in <i>RS</i> (million \$)	−2.22	−3.28
Percentage change in <i>RS</i>	−4.67	−6.87
Percentage change in <i>AS</i>	−4.66	−7.10
Change in <i>PS</i> (million \$)	4.23	4.32
Percentage change in <i>PS</i>	36.41	37.25

Welfare Effects of Ownership Consolidation

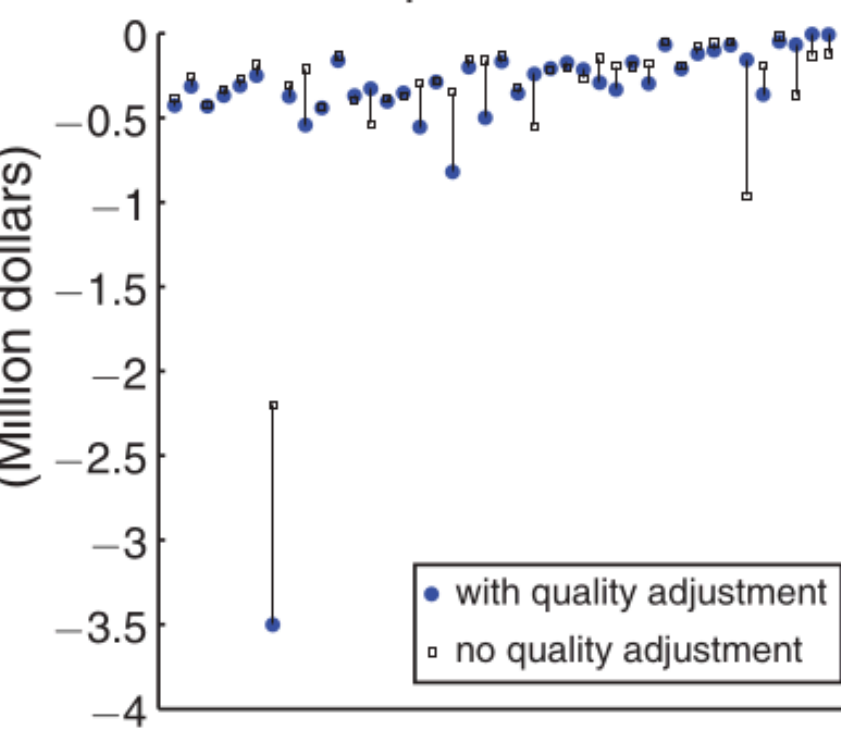
Underestimated welfare loss for consumers and gain for firms if we ignore endogenous product characteristics.

Duopoly Mergers

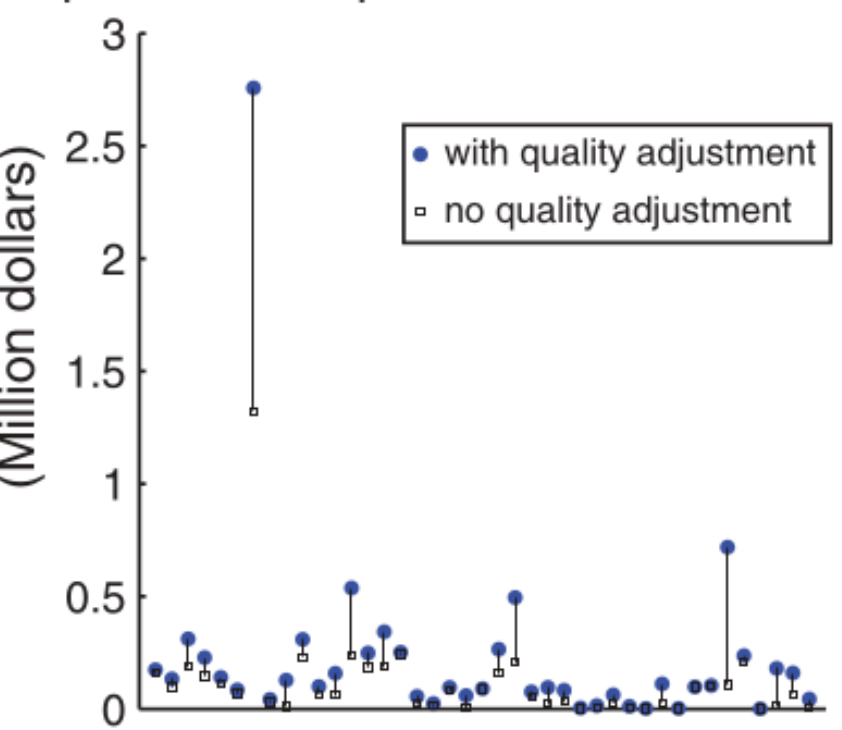
Panel A. Change in average per-household reader surplus



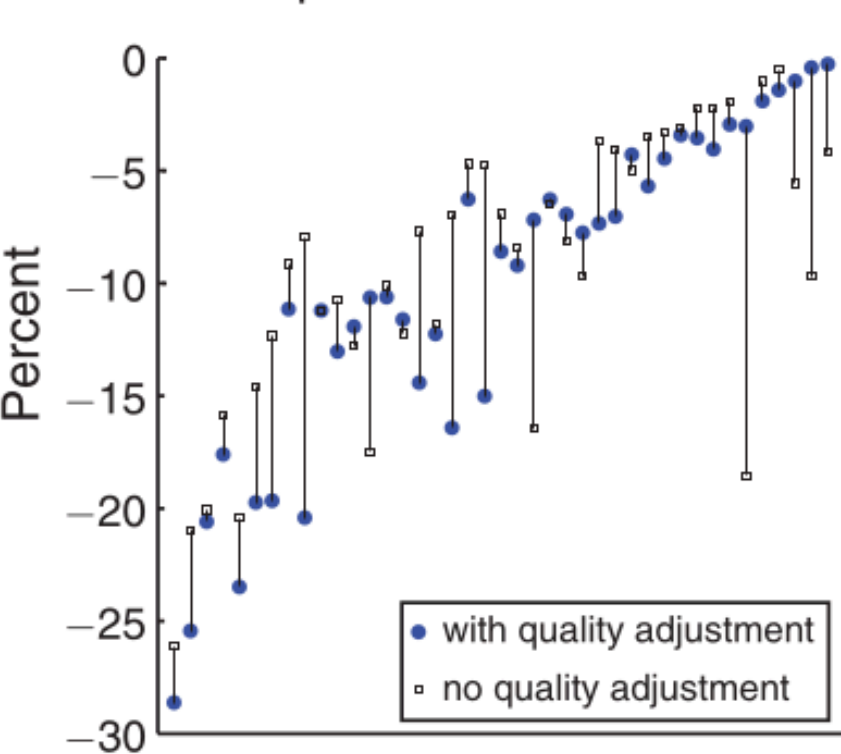
Panel B. Change in total reader surplus



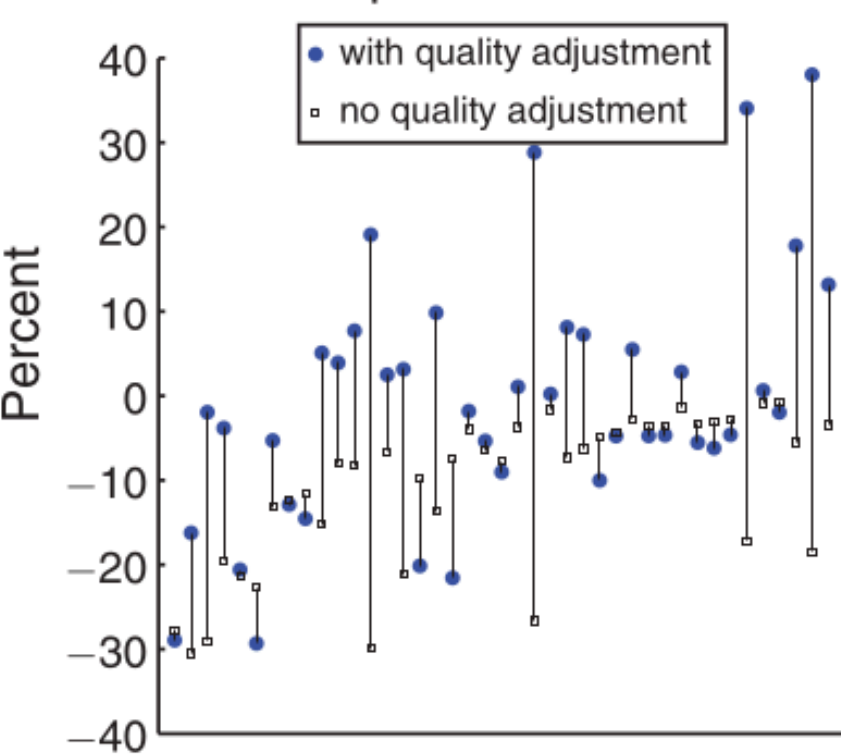
Panel C. Change in publisher surplus



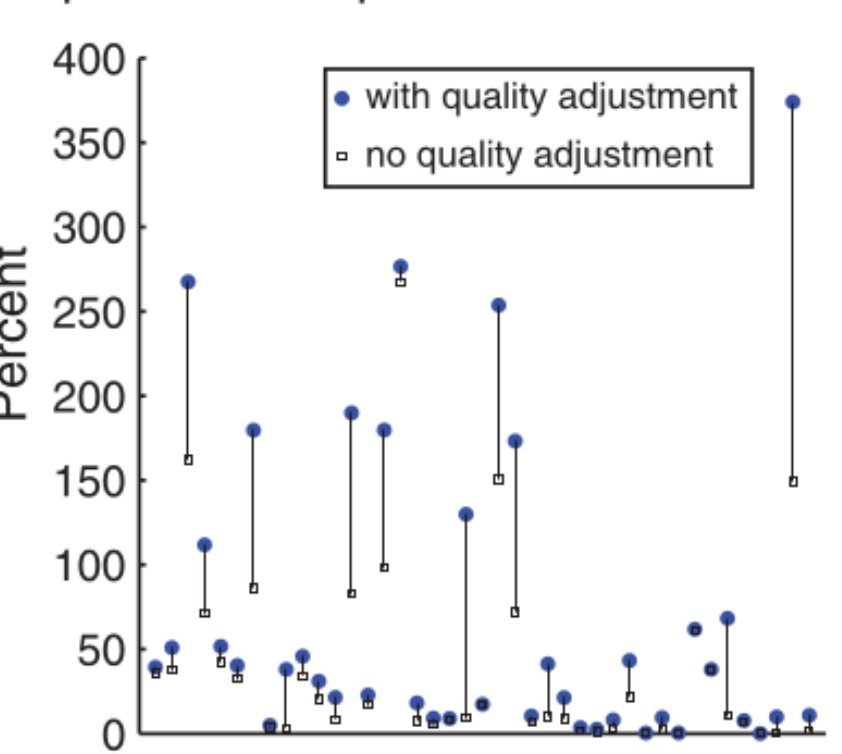
Panel D. Percentage change in reader surplus



Panel E. Percentage change in advertiser surplus



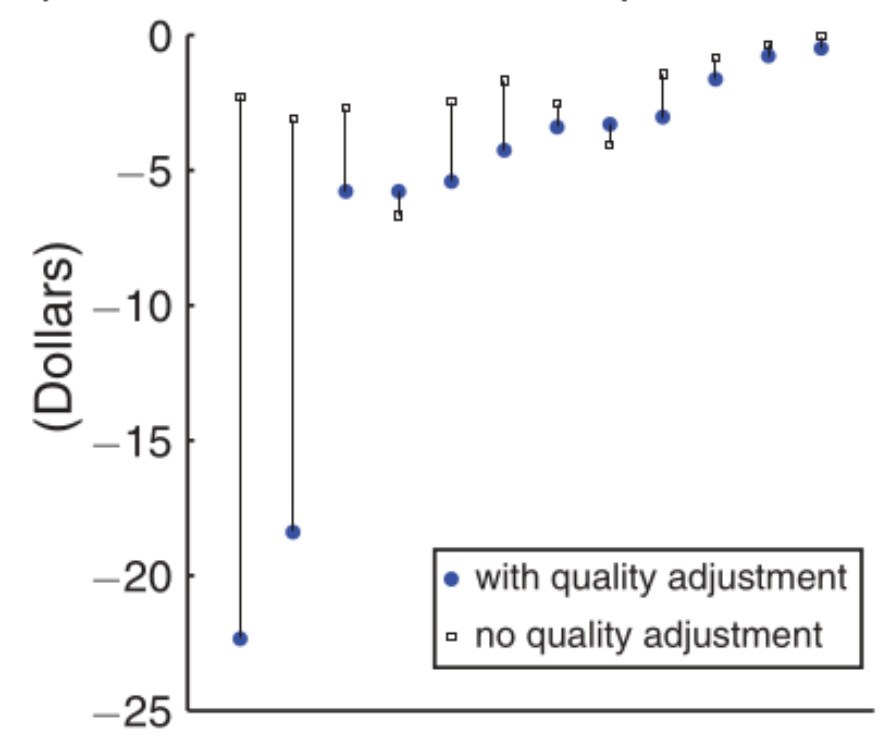
Panel F. Percentage change in publisher surplus



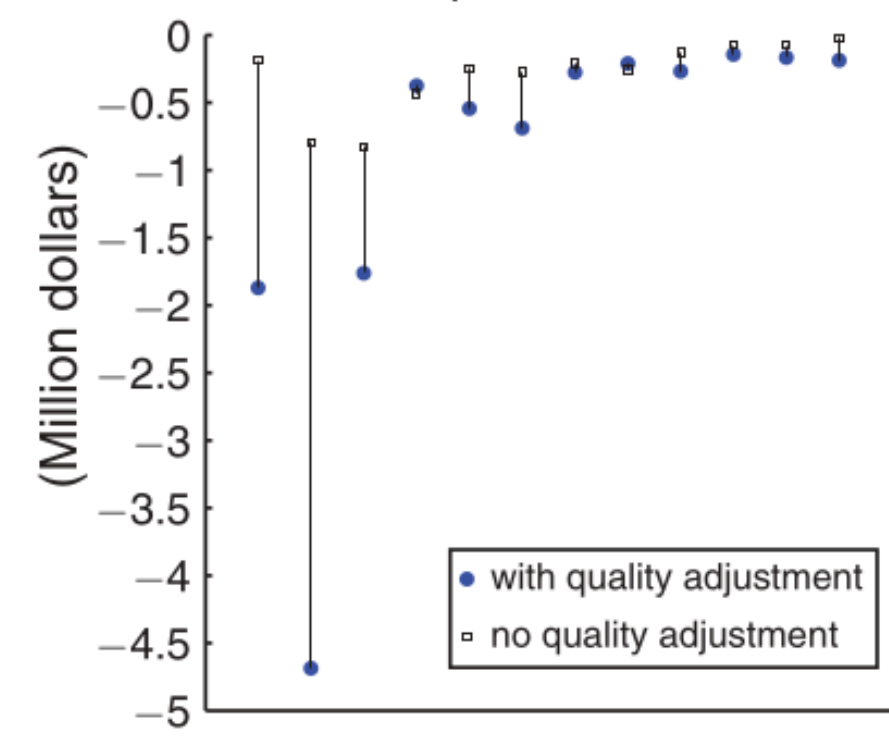
Welfare Implications of Duopoly Mergers

Triopoly Mergers

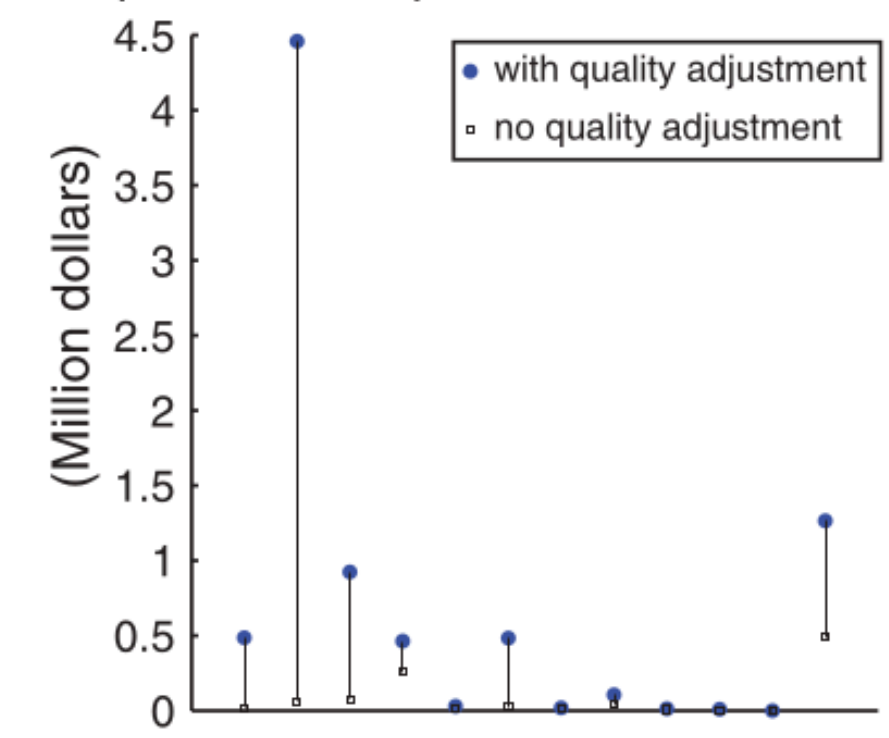
Panel A. Change in average per-household reader surplus



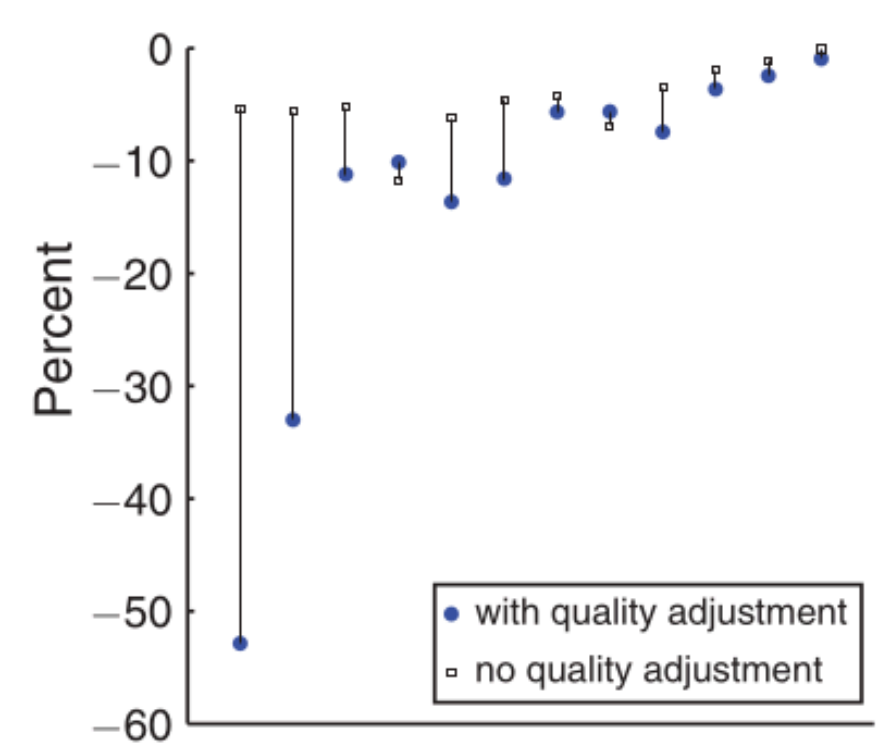
Panel B. Change in total reader surplus



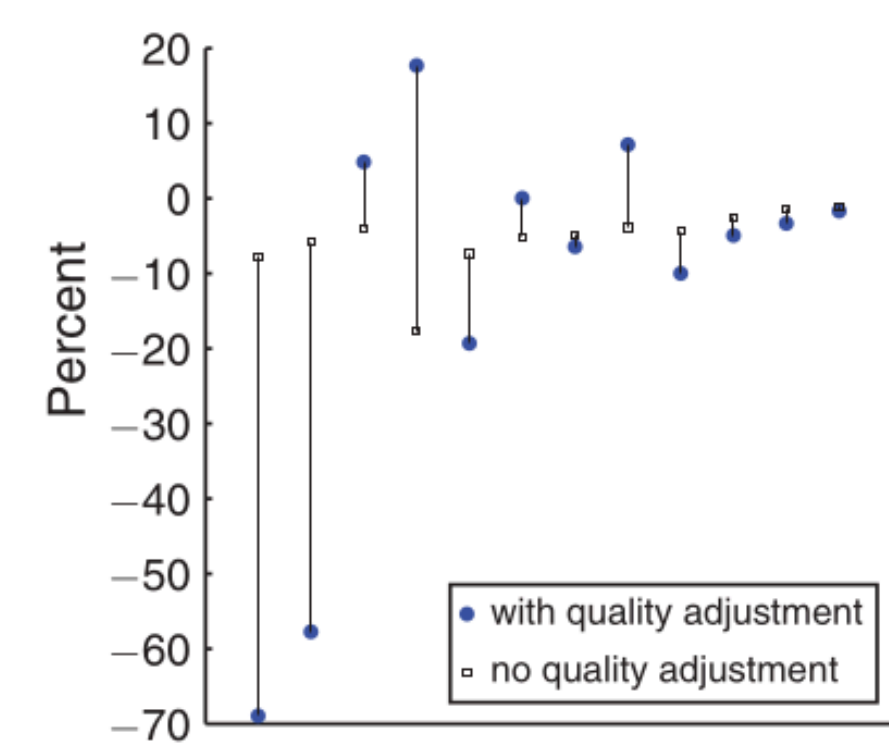
Panel C. Change in publisher surplus



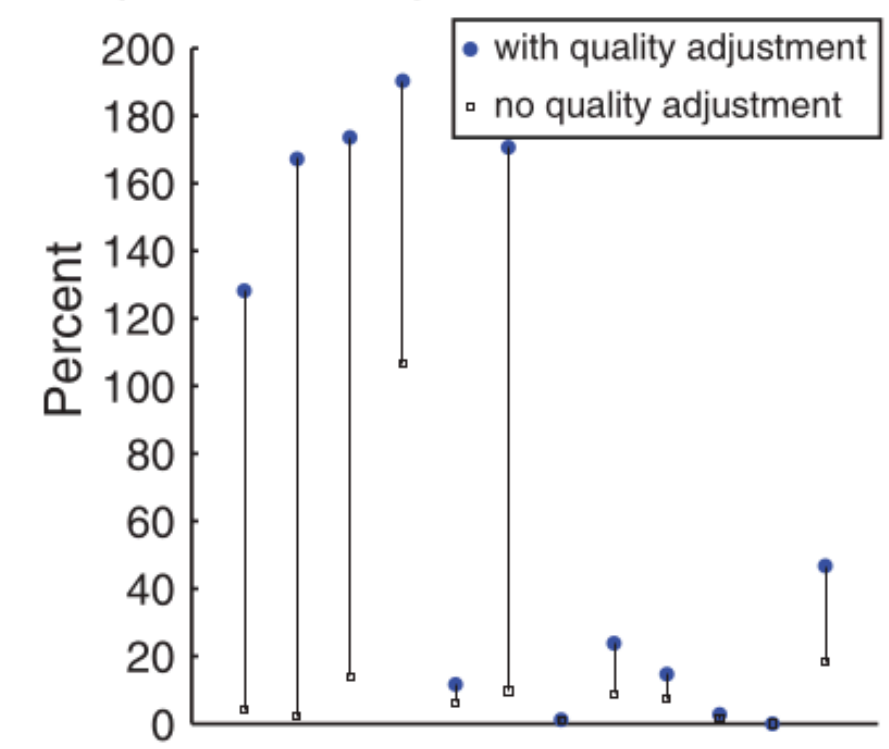
Panel D. Percentage change in reader surplus



Panel E. Percentage change in advertiser surplus



Panel F. Percentage change in publisher surplus



Welfare Implications of Triopoly Mergers

Conclusion

- Omitting product characteristics in merger analysis underestimates welfare loss for consumers and gain for firms
- Negative changes in reader's surplus due to mergers are in markets with
 - higher newspaper penetration
 - higher overlap of merged newspapers prior to the merger
 - comparable merged newspapers's sizes
 - duopoly market structure